Khang Tran

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Education

Princeton University, BA in Physics; minors in Computer Science and Engineering – Princeton, NJ — Sept 2024 – May 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Fixed Income, Options & Derivatives / Stochastic Calculus, Financial Risk Management (Fall 2026), Honors Linear Algebra, Multivariable Calculus, Complex Analysis (Fall 2026), Machine Learning & Pattern Recognition, Algorithms & Data Structures

NUS High School of Mathematics and Science, Diploma (High Distinction) with Honours in Engineering & Computer Science, and Majors in Mathematics, Chemistry & Physics – Singapore — Jan 2020 – Oct 2023

- **GPA:** 4.9/5.0 — Statistics, Advanced Statistics; Computer Organization, Operating Systems (NUS dual enrollment)

Honors and Competitions

- **Pyka Prize for Physics Excellence**, Princeton Physics Dept. (2026); Singapore Physics Olympiad **Gold** (2022); Singapore Junior Physics Olympiad **Gold** (2021); Singapore Physics League (SPhL) **Gold** (2022 — 8th place, 2023 — 3rd place)
- **COSCON 2025 1st** (Undergraduate); COSCON 2024 5th (Undergraduate)
- IBM Qiskit Hackathon **Champion** (Hardware track)

Experience

Undergraduate Researcher, Computational Physics, Roberto Car Group, Princeton University – Princeton, NJ — June 2026 – present

- Built a fault-tolerant, auto-relaunching pipeline to run large-scale numerical (DFT) computations across multi-node HPC (SLURM) clusters, automatically recovering from job failures to generate model training datasets.

Research Intern, Climate Data Analysis, High Meadows Environmental Institute, Princeton University – Princeton, NJ — June 2025 – Aug 2025

- Built Python pipelines to analyze large multi-model time-series ensembles, quantifying how varying forcing strengths drive a nonlinear climate response.

Space Systems Engineer Intern, Aliena – Singapore — Oct 2023 – Mar 2024

- Built an end-to-end Python automation pipeline for real-time control and data acquisition from laboratory instrumentation during hardware ignition tests.

Projects

Shielding-Free Signal-Noise Suppression in Portable Low-Field MRI – Singapore — Apr 2022 – Aug 2023

Singapore University of Technology and Design (SUTD)

- Built a signal-processing system to suppress noise in a portable MRI (advised by Prof. Huang Shaoying); **Silver**, Singapore Science & Engineering Fair 2023, and poster at ICMRM 2023.

Leadership and Teaching

- **Event Organizer**, Princeton Science Olympiad (2025–2026) & **Problem Setter**, SPhL (2023–2025); **Physics Tutor**, Princeton Physics Dept. (2025–2026) & **CS Lab TA**, COS217/226 (2026).

Skills

Programming: C/C++, Python (NumPy, Pandas, PyTorch, TensorFlow, scikit-learn), SQL, MATLAB

Quantitative: Probability, statistics, stochastic processes, numerical methods, statistical/ML modeling